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Computer vision-based assessment of building facade vulnerability and debris fall risk zones on the street during seismic events using street-level photography

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Abstract Summary : Street View Images (SVIs) have emerged as an increasingly valuable resource for comprehensive urban research applications. In seismically active regions, effective territorial management is essential for identifying structural weaknesses and enhancing urban resilience against earthquake hazards. Conventional seismic hazard evaluation approaches typically demand extensive on-site investigations and specialized instruments, creating significant barriers to widespread urban safety assessment due to resource limitations and the complexities of the process. The research introduces a methodology utilizing readily accessible SVIs data to evaluate pedestrian safety zones within urban environments, directly supporting adaptation strategies for hazardous geological events. Through the utilization of the sequential and multi-perspective characteristics of SVIs, image sequences can be reconstructed despite considerably large image intervals, enabling three-dimensional analysis for identifying hazardous and secure street zones using solely available visual data. This framework supports disaster-resilient infrastructure development and incorporates sustainable building enhancement strategies. The technique allows for comprehensive emergency preparedness encompassing risk reduction, damage limitation and, crisis response, through accessible evaluation tools. By utilizing existing urban documentation of Street View Images, actions can be taken using the proposed approach to address present and anticipated seismic hazards while developing actionable strategies that correspond with community needs for sustainable metropolitan growth. Through identification of secure pedestrian areas, findings would enable analysis of the risk locations during and after earthquakes, establishing safety evaluation as a fundamental component of municipal planning methodology. The risk assessment capabilities allow planners to pinpoint vulnerable districts and establish priorities for infrastructure enhancement, facilitating data-driven choices for seismic occurrences. This approach converts disaster resilience goals into municipal implementation by equipping city officials with economical instruments for thorough hazard mapping utilizing accessible street photography resources. The methodology supports evacuation pathway development, emergency protocol formulation, and regulatory frameworks considering seismic debris distribution patterns. Stakeholders and planners can employ these forecasts to direct building standard implementation, identify priority locations for structural evaluation, and guide public area development that strengthens community robustness. The system encourages ecological solution integration by locating areas where environmental infrastructure could offer additional safeguarding while preserving urban operations. Additionally, this assessment framework enables the surveillance and revision of hazard maps as landscapes transformation data for consideration, ensuring planning choices remain relevant and adaptive to evolving circumstances. The availability of street imagery makes this approach especially beneficial for data-limited places seeking comprehensive disaster reduction implementation without extensive requirements, ultimately fostering more resilient and responsive urban ecosystems.